

Project Details (FEnd Only)

Dashboard GrowTH KKU intelsphere

https://gen.ai.kku.ac.th/webdev/019f2636-42ed-7e26-9ffa-1c9e7d72f071?fbclid=IwY2xjawS0TxRleHRuA2FibQlzMABicmlkETFqTVJSUIh6Wk5TalplbXQ1c3J0YwZhcHBfaWQQMjlyMDM5MTc4ODlwMDg5MgABHuiFVc1T3X_sKNWN7ZdrccE2VjJ_HILsJ0vA45CaBdegLR6nCBVINmQKtfS_aem_vFSpONBDjP2R-0CwoYjTka

Cloud hosting

- **Vercel** - Auto-deploys from GitHub, gives a stable URL, very beginner-friendly, auto-detects React projects

Language

- HTML/CSS, JavaScript

Framework

- React

2. Objectives

The objectives of this project are as follows:

1. To design and develop a user-friendly web application (GrowTH) that allows parents to register their children, record growth measurements (height and weight), and view automatically generated growth analysis, including percentile position and standard deviation score (SDS), based on recognized pediatric growth references.
2. To design and develop a structured puberty development screening module within the same application, allowing parents to answer a guided set of questions about physical and developmental signs and receive a preliminary screening result.
3. To design, train, and integrate a machine learning model capable of estimating bone age from an uploaded hand-and-wrist X-ray image, using the RSNA Pediatric Bone Age dataset as the primary training and validation data source, and to make the resulting estimate available within the application as a supporting input to the puberty and growth screening process.
4. To design a complete and consistent user experience and user interface (UX/UI) in Figma covering all in-scope features of the application, prior to development.
5. To build a secure backend system and database capable of storing user accounts, child profiles, growth records, screening responses, and bone age prediction results, designed with appropriate consideration for the sensitivity of children's health data.
6. To produce a coordinated set of promotional video content — comprising a doctor interview, a 2D motion graphic narrative, an application promotional video, an application demonstration video, and a set of short-form video clips — to support the public launch and awareness campaign for GrowTH.
7. To deliver a functioning, demonstrable version of the application and the full set of promotional video assets by the final launch date of November 2.

2A. Assumptions, Target Users, and Constraints

2A.1 Target Users

GrowTH is designed primarily for the following user groups:

- Parents and primary caregivers of children from infancy through adolescence, who wish to track their child's physical growth on an ongoing basis between clinic visits.
- Parents and caregivers who have noticed, or wish to proactively screen for, early signs of pubertal development that may fall outside the typical age range for their child's sex.
- Secondary audience for the promotional video campaign: the broader public, including extended family members and educators, who may benefit from increased awareness of puberty development disorders and may share or recommend the application to parents.

2A.2 Key Assumptions

- The project team will have access to a standard set of pediatric growth reference data (e.g., WHO or national growth standards) suitable for calculating percentiles and SDS values; the specific reference dataset to be used shall be confirmed with the Client Representative early in the project.
- The RSNA Pediatric Bone Age dataset is assumed to be available for download and use under its existing public license for the purposes of model training and academic evaluation.
- The doctor interview video will feature a role-played physician character rather than an interview with an actual practicing physician. The Project Team is responsible for casting a presenter for this role and ensuring the role-player is appropriately briefed. The Client Representative will review and approve the interview script/talking points for clinical accuracy prior to filming, but is not responsible for arranging a real physician's availability.

2A.3 Constraints

- This TOR does not allocate a project budget; the project team shall design and build the solution using tools, datasets, and infrastructure that are free, open-source, or otherwise already available to the team and the institution.
- This TOR does not prescribe a detailed phase-by-phase schedule; the only fixed date is the Final Launch Date of November 2, described in Section 10.
- The application is a class project deliverable and is not intended for clinical deployment or commercial release without further regulatory, clinical-validation, and security review beyond the scope defined here.

3.2 Web Application Development (Front End)

The project team shall develop the responsive front-end of the GrowTH web application based on the approved Figma design, implementing all functional modules described in Section 4 (Functional Requirements). Front-end development activities include:

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- Implementation of all approved screens and user flows as functioning, responsive web pages compatible with current major desktop and mobile browsers.
 - Implementation of data visualization components, including growth charts (height-for-age, weight-for-age, BMI-for-age) with the child's current data point plotted against standard percentile curves.
 - Implementation of form-based data entry screens with appropriate input validation (e.g., valid date ranges, plausible height/weight values, required fields) and clear, parent-friendly error messaging.
 - Implementation of the bone age X-ray upload interface, including file type/size validation, upload progress indication, and clear presentation of the returned AI prediction result alongside an appropriate explanatory note on how to interpret it.
 - Integration with the backend API for all data read/write operations described in Section 3.3.
 - Basic accessibility considerations (readable font sizes, sufficient color contrast, descriptive form labels) appropriate for a parent-facing health application
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4. Functional Requirements

This section describes the functional requirements of the Growth web application in further detail, organized by module. A consolidated summary table is also provided in Annex A.

4.1 Account and Profile Management

- FR-1: The system shall allow a new user (parent/caregiver) to register an account using, at minimum, full name, email address, password, and phone number.
- FR-2: The system shall require the user to accept the application's terms of use and privacy notice before account creation is completed.
- FR-3: The system shall allow a registered user to log in and log out securely.
- FR-4: The system shall allow a user to add one or more child profiles, capturing at minimum: child's name, sex, date of birth, and relationship to the user.
- FR-5: The system shall allow a user to view, edit, and switch between multiple child profiles linked to their account.

Rationale: Many households include more than one child, and a parent should be able to manage growth and screening records for each child independently within a single account, without needing to register separately for each child.

4.2 Growth Tracking and Screening

- FR-6: The system shall allow a user to record a growth entry for a selected child, including date of measurement, height, and weight.
- FR-7: The system shall calculate and display the child's percentile and standard deviation score (SDS) for height-for-age and weight-for-age based on an agreed standard growth reference.
- FR-8: For children aged five years and above, the system shall calculate and display BMI and BMI-for-age SDS, with an accompanying plain-language nutritional status summary (e.g., "within normal range").
- FR-9: The system shall display growth trends over time using line charts for height-for-age, weight-for-age, and BMI-for-age, plotted against standard

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- FR-9: The system shall display growth trends over time using line charts for height-for-age, weight-for-age, and BMI-for-age, plotted against standard percentile curves.
- FR-10: The system shall provide a plain-language summary and basic guidance message following each growth entry (e.g., flagging when a result falls notably outside the expected range and recommending consultation with a physician), without presenting this guidance as a clinical diagnosis.
- FR-11: The system shall allow the user to view a complete history of past growth entries for a given child.

Rationale: Growth assessment is inherently longitudinal — a single measurement is far less informative than a trend over time. The historical view and trend charts are therefore considered core, not optional, functionality.

4.3 Puberty Development Screening

- FR-12: The system shall provide a structured, guided questionnaire allowing a parent to report observed signs of pubertal development appropriate to the child's sex and age.
- FR-13: The system shall compile the questionnaire responses into a preliminary screening result, clearly labeled as a screening aid rather than a diagnosis, with guidance on recommended next steps (e.g., consulting a pediatrician or pediatric endocrinologist).
- FR-14: The system shall allow the user to view a history of previously completed puberty screening assessments for a given child.

Rationale: A single screening response is a snapshot; the ability to see how responses change over successive assessments helps a parent and, eventually, a treating physician understand the trajectory of the child's development rather than a single point in time.

4.4 AI-Assisted Bone Age Prediction

- FR-15: The system shall allow a user to upload a hand-and-wrist X-ray image file (in a defined set of accepted formats) for a selected child.
 - FR-16: The system shall validate the uploaded file for format and file size, and shall provide clear feedback if the upload does not meet requirements.
 - FR-17: The system shall submit the uploaded image to the bone age prediction model (Section 3.4) and display the predicted bone age result to the user within a reasonable processing time.
 - FR-18: The system shall present the bone age result together with an explanatory note describing what the figure means, its margin of error, and that it is intended to support — not replace — clinical assessment.
 - FR-19: The system shall associate each bone age prediction result with the corresponding child profile and growth/screening history, allowing the parent to review past predictions over time.
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prediction model (Section 3.4) and display the predicted bone age result to the user within a reasonable processing time.

- FR-18: The system shall present the bone age result together with an explanatory note describing what the figure means, its margin of error, and that it is intended to support — not replace — clinical assessment.
- FR-19: The system shall associate each bone age prediction result with the corresponding child profile and growth/screening history, allowing the parent to review past predictions over time.

Rationale: The bone age feature is most clinically meaningful when considered alongside the child's growth and puberty screening history rather than in isolation; linking these data points within the application supports a more complete picture for both the parent and any physician the parent later consults.

4.5 Knowledge / Education Section

- FR-20: The system shall provide a knowledge section containing reference articles and/or reference growth chart materials relevant to child growth and

puberty development, sourced or adapted with appropriate attribution and without infringing copyright.

- FR-21: Content in the knowledge section shall be organized and searchable/browsable by topic.

4.6 General / Non-Functional Considerations Relevant to Functionality

- FR-22: The system shall be usable on both desktop and mobile web browsers through a responsive layout.
- FR-23: The system shall display all clinical or quasi-clinical results (growth status, screening result, bone age prediction) using plain, parent-friendly

- FR-21: Content in the knowledge section shall be organized and searchable/browsable by topic.

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- FR-22: The system shall be usable on both desktop and mobile web browsers through a responsive layout.
- FR-23: The system shall display all clinical or quasi-clinical results (growth status, screening result, bone age prediction) using plain, parent-friendly language, supplemented with the underlying figures for users who want more detail.
- FR-24: The system shall ensure that a given user can only access the profiles and data of children linked to their own account.

6. Technical Requirements and Standards

6.1 Platform and Compatibility

- The application shall be delivered as a responsive web application, accessible via current versions of major web browsers (e.g., Chrome, Safari, Edge, Firefox) on both desktop and mobile devices.
- The application shall be hosted in an environment accessible via a stable URL for the duration of the evaluation and demonstration period.
- The project team shall select and justify their technology stack (front-end framework, backend framework/language, database system, and model-serving approach) in the system overview document (Deliverable D10), considering maintainability, suitability for the dataset and model type, and feasibility within the project timeline.

6.2 Data Privacy and Protection

Given that GrowTH collects and processes health-related information about children, which is sensitive personal data, the project team shall design the system with the following principles in mind, consistent with the spirit of Thailand's Personal Data Protection Act (PDPA):

- Collection of personal data shall be limited to what is necessary for the application's stated functions (data minimization).
 - The application shall present a clear privacy notice explaining what data is collected, how it is used, and how it is stored, prior to account registration.
 - Uploaded X-ray images and associated prediction results shall be stored securely and shall be accessible only to the account that uploaded them.
 - Passwords shall never be stored in plain text; an industry-standard hashing approach shall be used.
 - The project team shall document, in the system overview document, what data is collected, where and how it is stored, and what (if any) data would need to be deleted upon a user's request, to demonstrate awareness of data subject rights even though full PDPA compliance certification is not required for this class project.
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7. Project Team Qualifications

As this project is conducted as a class assignment for third-year students of the Digital Media Engineering program, qualifications are framed around relevant coursework, demonstrated skills, and team composition rather than professional work history. The project team is expected to be a multidisciplinary group of students collectively able to cover the following skill areas:

- UX/UI Design: Working proficiency in Figma, including component-based design systems, responsive layout principles, and prototyping; familiarity with basic usability evaluation methods.
- Front-End Development: Working knowledge of a modern web front-end framework (e.g., React, Vue, or equivalent), responsive CSS techniques, and consumption of REST or equivalent APIs.

9.2 Milestone Reviews

In addition to the weekly update meeting, the following milestone reviews shall be scheduled at the relevant points in the project, each requiring formal sign-off from the Client Representative before the team proceeds to the next phase of work:

Milestone	Review Focus
M1 – UX/UI Design Review	Review of wireframes and high-fidelity Figma designs and prototype against the requirements in Sections 3.1 and 4
M2 – Application Beta 1 Review	Review of a working but incomplete build of the front end and backend, demonstrating core data flows (account creation, growth entry, screening flow)
M3 – AI Model Review	Review of the trained bone age model's evaluation results (including MAE) and a demonstration of the integrated prediction flow within the application
M4 – Video Script/Storyboard Review	Review and approval of scripts and/or storyboards for all five video categories described in Section 3.5, prior to filming/animation production
M5 – Application Beta 2 Review	Review of a feature-complete application build, including all functional requirements in Section 4, for user acceptance testing
M6 – Video Rough-Cut Review	Review of rough-cut versions of all video deliverables for feedback prior to final editing

Annex C: Glossary of Terms

Term	Definition
Bone Age	An estimate of skeletal maturity derived from the appearance of bones (typically of the hand and wrist) on an X-ray, compared against population reference standards.
Precocious Puberty	The onset of pubertal physical changes at an earlier age than typical, which can be associated with accelerated bone maturation and reduced final adult height if untreated.
MAE (Mean Absolute Error)	A model evaluation metric representing the average absolute difference between predicted and actual values; in this project, expressed in months of bone age.
Percentile / SDS	Statistical measures used to express how a child's height, weight, or BMI compares to a reference population of children of the same age and sex; SDS stands for Standard Deviation Score (also known as a z-score).
RSNA Pediatric Bone Age Dataset	A publicly available dataset of labeled hand-and-wrist X-ray images, compiled by the Radiological Society of North America, widely used for training and benchmarking automated bone age prediction models.
PDPA	Thailand's Personal Data Protection Act, which sets requirements for the lawful collection, use, and protection of personal data, including health-related data.
Figma	A cloud-based collaborative interface design tool used for creating wireframes, high-fidelity mockups, and interactive prototypes.
UX/UI	User Experience and User Interface design — respectively, the overall usability and flow of an application, and its visual presentation and interactive elements.

Annex D: Indicative Puberty Screening Question Categories

The following question categories are provided as an indicative starting point for the puberty development screening questionnaire (FR-12). The exact wording, clinical thresholds, and scoring logic shall be finalized in consultation with the Client Representative and, where appropriate, reviewed against established clinical screening references before implementation.

D.1 For Female Children

- Has breast development (thelarche) been observed, and at approximately what age?
- Has pubic or underarm hair growth been observed, and at approximately what age?
- Has menstruation begun, and at approximately what age?
- Has the child experienced a noticeable recent increase in height growth rate?

D.2 For Male Children

- Has testicular or genital enlargement been observed, and at approximately what age?
- Has pubic, underarm, or facial hair growth been observed, and at approximately what age?
- Has voice deepening been observed?
- Has the child experienced a noticeable recent increase in height growth rate?

D.2 For Male Children

- Has testicular or genital enlargement been observed, and at approximately what age?
- Has pubic, underarm, or facial hair growth been observed, and at approximately what age?
- Has voice deepening been observed?
- Has the child experienced a noticeable recent increase in height growth rate?

D.3 General Questions (Both Sexes)

- Family history: at approximately what age did the child's parents or siblings begin puberty?
- Has the child shown any behavioral, mood, or skin changes (e.g., acne, body odor) that the parent associates with early physical development?
- Are there any other health conditions, medications, or relevant medical history the parent wishes to record?

The result of this questionnaire shall be compiled into a plain-language screening summary, as described under FR-13, and shall not be presented as a clinical diagnosis.

Annex A: Functional Requirement Summary Table

ID	Requirement Summary	Module
FR-1 to FR-5	Account registration, login, terms acceptance, child profile management	Account & Profile
FR-6 to FR-11	Growth entry, percentile/SDS calculation, BMI-for-age, trend charts, guidance messages, history view	Growth Tracking
FR-12 to FR-14	Guided puberty screening questionnaire, preliminary result, history view	Puberty Screening
FR-15 to FR-19	X-ray upload, validation, AI bone age prediction, result explanation, history view	AI Bone Age
FR-20 to FR-21	Knowledge section content and organization	Knowledge / Education
FR-22 to FR-24	Responsive design, plain-language results, account-scoped data access	General / Non-Functional

To Do List

03/07/2026 - 05/07/2026

1. Learn

- **JavaScript basics** — variables, functions, arrays/objects, and especially `fetch`/async-await (how your app will talk to the backend later)
- **React basics** — components, props, `useState`, `useEffect`
- Free resources: react.dev's official tutorial, or freeCodeCamp's React course on YouTube

2. Set up your project structure (Task 1 — you can start this now)

This task doesn't depend on final designs at all:

- Install Node.js if you haven't
- Run `npm create vite@latest` to scaffold your React project
- Set up your GitHub repo for the front-end code
- Connect the repo to Vercel (so you have a working stable URL from day one, even if it just shows a blank placeholder page)
- Install and test Chart.js/Recharts with fake/dummy data, just to confirm it works before real data exists

3. Plan your Component Requirements based on the TOR (not final designs)

You already know *what* screens are needed from Section 4 of the TOR, even without exact visuals:

- Login/Register screen
- Child profile list + add/edit form
- Growth entry form + chart display
- Puberty screening questionnaire
- Bone-age X-ray upload screen
- Knowledge/articles section

You can sketch out a rough **folder/component structure** for these now, so once designs arrive you're just filling in styling, not starting from zero.

src/

├── components/

| ├── Auth/

| | ├── LoginForm.jsx

| | └── RegisterForm.jsx

| ├── ChildProfile/

| | ├── ChildProfileList.jsx

| | └── ChildProfileForm.jsx

| ├── GrowthTracking/

| | ├── GrowthEntryForm.jsx

| | └── GrowthChart.jsx

| ├── PubertyScreening/

| | └── PubertyQuestionnaire.jsx

| ├── BoneAge/

| | └── XrayUploadForm.jsx

| └── Knowledge/

| └── ArticleList.jsx

├── pages/

| ├── LoginPage.jsx

| ├── DashboardPage.jsx

| ├── GrowthPage.jsx

| ├── PubertyPage.jsx

| └── BoneAgePage.jsx

| └── KnowledgePage.jsx

└── App.jsx

4. Build with placeholder/dummy data

Practice building basic versions of components (e.g., a simple form, a basic line chart with fake numbers) using generic styling. This isn't wasted work — once real designs come in, you restyle these components rather than rebuild the logic from scratch.

5. Coordinate with your team

- Ask UX/UI teammates for a rough timeline on when Figma designs will be ready
- Ask backend teammates what the API structure/endpoints will roughly look like, so you know what data shape to expect (helps you plan components correctly)

Component Requirments

GrowTH Front-End — Component Requirements Plan

This document lists what each planned component needs to contain, based directly on the TOR's Functional Requirements (Section 4). Use this as a reference while building placeholder components, before final Figma designs arrive.

1. Auth / Login & Register (FR-1, FR-2, FR-3)

RegisterForm.jsx

- Full name input
- Email address input
- Password input
- Phone number input
- Checkbox: "I accept the terms of use and privacy notice" (required — cannot submit without checking)
- Submit button
- Link to switch to Login

LoginForm.jsx

- Email input
 - Password input
 - Submit (login) button
 - Logout action (can live in a nav/header component instead)
-

2. Child Profile Management (FR-4, FR-5)

ChildProfileForm.jsx (add/edit)

- Child's name input
- Sex selector (e.g., male/female)
- Date of birth picker
- Relationship to user input (e.g., parent, guardian)

- Submit button

ChildProfileList.jsx

- List/grid of existing child profiles
 - Each item: name, photo/avatar placeholder, quick view button
 - "Switch active child" action
 - "Add new child" button
 - Edit button per profile
-

3. Growth Tracking & Screening (FR-6 to FR-11)

GrowthEntryForm.jsx

- Date of measurement (date picker)
- Height input (numeric, needs validation — plausible range)
- Weight input (numeric, needs validation — plausible range)
- Submit button
- Error messages for invalid/out-of-range values

GrowthChart.jsx

- Line chart: height-for-age plotted against standard percentile curves
- Line chart: weight-for-age plotted against standard percentile curves
- Line chart: BMI-for-age (only shown for children aged 5+, per FR-8)
- Display of current percentile + SDS (standard deviation score) as text/number, not just the chart
- Plain-language nutritional status summary (e.g., "within normal range") for BMI, ages 5+

GrowthGuidanceMessage.jsx

- Plain-language summary shown after each entry (FR-10)
- Flag/highlight if a result is notably outside expected range
- Recommendation text to consult a physician when flagged
- Must NOT present this as a clinical diagnosis (just wording/tone note for content, not really a "field")

GrowthHistoryList.jsx

- Table/list of all past growth entries for the selected child (FR-11)
- Columns: date, height, weight, calculated percentile/SDS

4. Puberty Development Screening (FR-12 to FR-14)

PubertyQuestionnaire.jsx

- A structured, guided set of questions (content will vary by child's sex and age — logic needed to show correct question set)
- Each question likely needs: question text + answer input (yes/no, multiple choice, or similar)
- Submit button

PubertyResult.jsx

- Preliminary screening result display
- Must be clearly labeled as a "screening aid," NOT a diagnosis
- Guidance text on recommended next steps (e.g., "consider consulting a pediatrician")

PubertyHistoryList.jsx

- History of previously completed puberty screening assessments for the selected child (FR-14)
- Shows date + result summary per past assessment

5. AI-Assisted Bone Age Prediction (FR-15 to FR-19)

XrayUploadForm.jsx

- File upload input (hand-and-wrist X-ray image)
- Accepted file format/size validation with clear error messaging if rejected (FR-16)
- Upload progress indicator
- Submit button

BoneAgeResult.jsx

- Display predicted bone age result (returned from backend/AI model)
- Explanatory note: what the number means, margin of error, and that it supports — not replaces — clinical assessment (FR-18)
- Link/reference back to the selected child's profile

BoneAgeHistoryList.jsx

- List of past bone age predictions for a given child (FR-19)
 - Shown alongside/linked to that child's growth and puberty screening history
-

6. Knowledge / Education Section (FR-20, FR-21)

ArticleList.jsx

- Browsable/searchable list of reference articles or growth chart materials
- Organized by topic
- Search or filter input (FR-21: "searchable/browsable by topic")

ArticleDetail.jsx

- Full article/content view
 - Source/attribution note (content must be properly attributed, not infringing copyright)
-

7. General / Cross-Cutting Requirements (FR-22 to FR-24)

These aren't single components, but rules that apply across ALL components above:

- **Responsive layout** — every screen must work on both desktop and mobile browsers (FR-22)
 - **Plain-language + detail toggle** — all clinical/quasi-clinical results (growth status, screening result, bone age) must show in simple parent-friendly language by default, with an option to reveal the underlying numeric figures for users who want more detail (FR-23)
 - **Access control** — a user must only ever see profiles/data for children linked to their own account (FR-24) — this is mostly a backend/API rule, but front-end should never display another user's data even if an API bug exposes it
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Quick Checklist

Component	Status
• RegisterForm.jsx	☐ Not started
• LoginForm.jsx	☐ Not started
• ChildProfileForm.jsx	☐ Not started
• ChildProfileList.jsx	☐ Not started
• GrowthEntryForm.jsx	☐ Not started
• GrowthChart.jsx	☐ Not started
• GrowthGuidanceMessage.jsx	☐ Not started
• GrowthHistoryList.jsx	☐ Not started
• PubertyQuestionnaire.jsx	☐ Not started
• PubertyResult.jsx	☐ Not started
• PubertyHistoryList.jsx	☐ Not started
• XrayUploadForm.jsx	☐ Not started
• BoneAgeResult.jsx	☐ Not started
• BoneAgeHistoryList.jsx	☐ Not started
• ArticleList.jsx	☐ Not started
• ArticleDetail.jsx	☐ Not started

Update this doc as UX/UI designs finalize and requirements are confirmed with the Client Representative.